

Grocery Management System



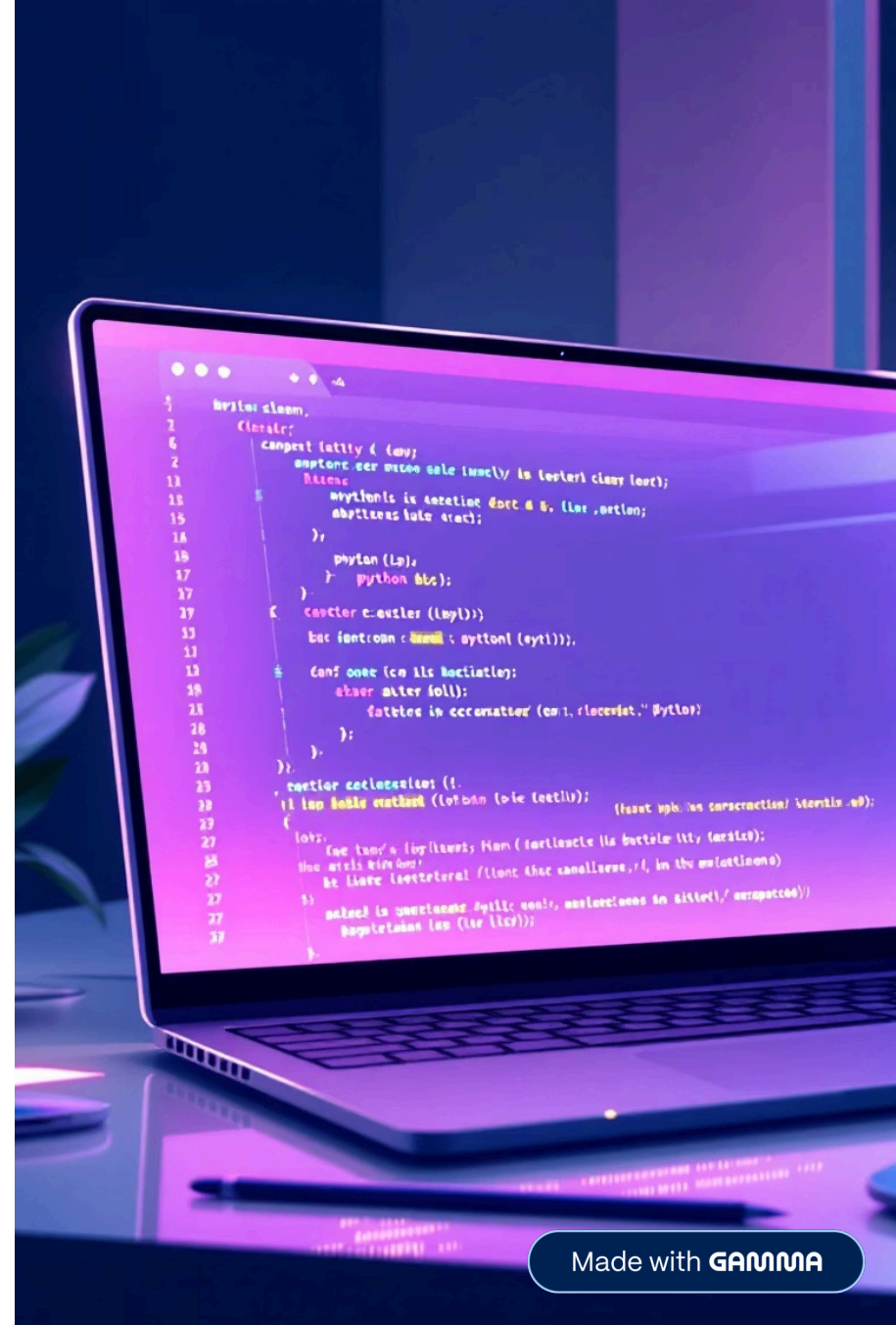
Project Overview

What It Does

A console-based application managing grocery items through core Python operations: adding, viewing, searching, editing, and purchasing items.

Core Approach

Built using Object-Oriented Programming with classes, objects, methods, and data encapsulation—no external libraries required.



Project Objectives



Usable System

Create an easily usable inventory management system for real-world application.



Demonstrate OOP

Showcase Python OOP concepts in practical implementation.



CRUD Operations

Manage grocery items using complete Create, Read, Update, Delete operations.



Practical Learning

Practice input handling, error checking, and data structure usage.

Technologies & Concepts

1

Core Python

Built entirely with Python—no external libraries needed.

2

OOP Principles

Classes, objects, and methods form the foundation.

3

Control Flow

While loops and conditional statements manage program logic.

4

Data Structures

Lists and dictionaries store and organize inventory data.

5

Error Handling

Exception handling ensures smooth operation and user experience.

System Features



Add Item

Add new grocery items with name, quantity, and price details.



Purchase Item

Buy items with automatic quantity reduction and price calculation.



Edit Item

Modify item name, quantity, or price as needed.



View Items

Display complete inventory with all item details at a glance.



Search Item

Find any item by name and view complete details instantly.



Exit

Safely close the program when operations are complete.

How It Works



Start Program

User launches the application



Display Menu

Main menu shows all options



User Choice

Select option 1-6



Execute Task

System performs operation



Loop or Exit

Menu repeats until exit



Class Structure

GroceryStore

The main class controlling all operations through organized methods and clean object structure.

```
store = GroceryStore()  
store.run()
```

Core Methods

- `view_items()` – Displays all items
- `add_item()` – Adds new item
- `purchase_item()` – Handles purchases
- `search_item()` – Searches by name
- `edit_item()` – Modifies details
- `run()` – Controls program flow

Sample Operations

Main Menu Display

1. View Items
2. Add Item
3. Purchase Item
4. Search Item
5. Edit Item
6. Exit

Adding an Item

Item name: Sugar
Item quantity: 20
Item price: 45
Item added successfully!

Searching Items

Enter item name: Sugar
Found: {'name': 'Sugar',
'quantity': 20, 'price': 45.0}

Purchase Process

Enter item name: Milk
Pay ₹50 at counter.
Item purchased!



Project Benefits & Future Vision

Key Advantages

- Simple and easy to operate
- Built with Core Python only
- Clear OOP concept demonstration
- Extendable to real store systems
- Perfect for academic projects

Future Enhancements

- Add SQL database support
- Generate billing receipts
- Implement admin/user login
- Add categories and filtering
- Create GUI with Tkinter

Thank You!

We appreciate your time and attention today..